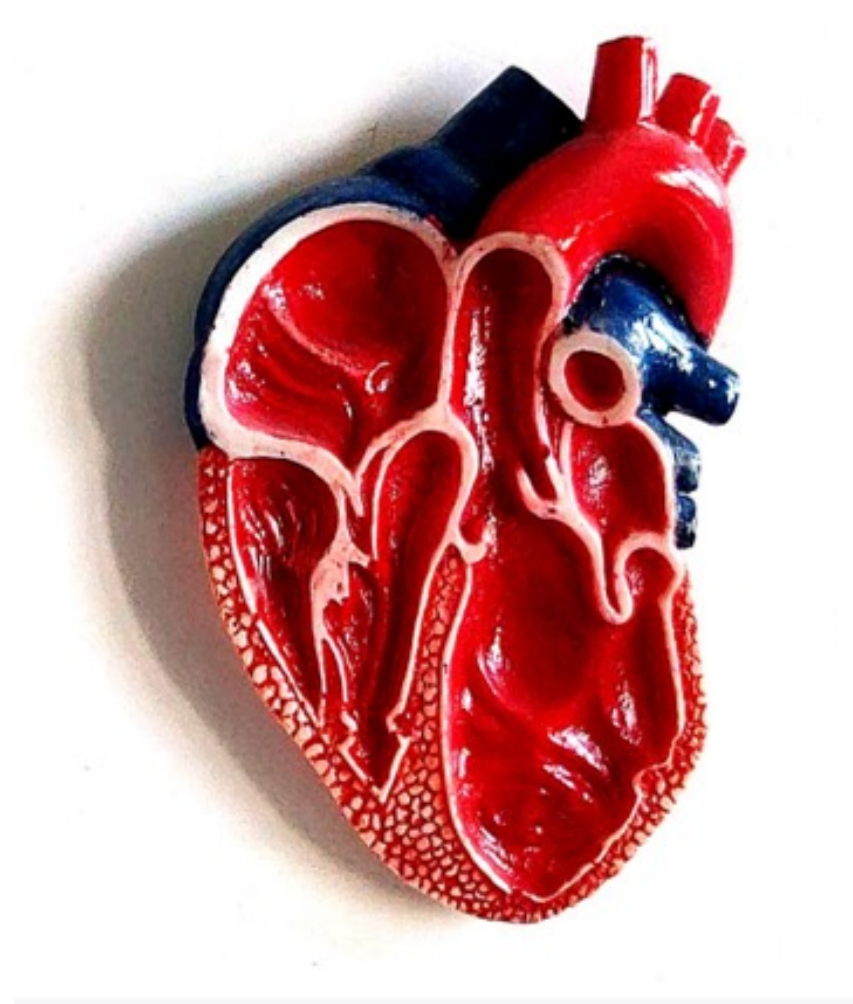


Heart failure

Clinical correlation IV

Nitchanun Jinpisoot, MD.



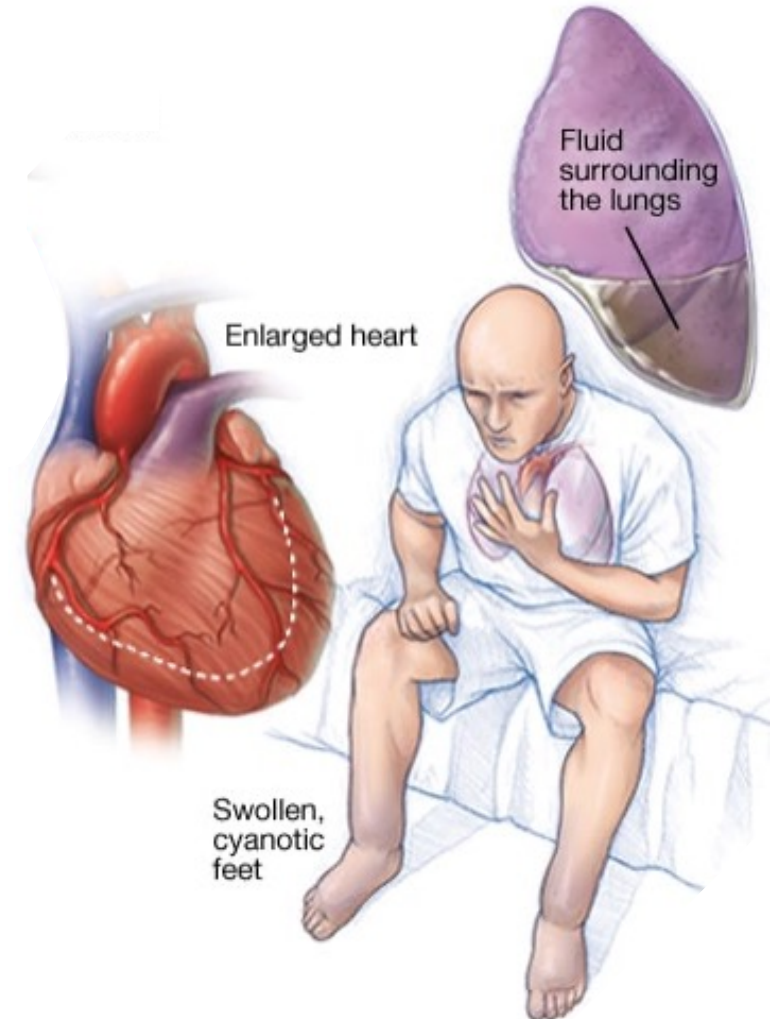
Outline

-
- Definition of heart failure
-
- Pathophysiology of heart failure
-
- Clinical features of heart failure
-
- How to diagnosis heart failure
-
- Treatment

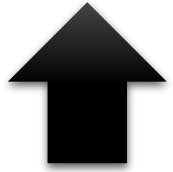
HF Description



Heart failure is a complex clinical syndrome with symptoms and signs that result from any structural or functional impairment of ventricular filling or ejection of blood.



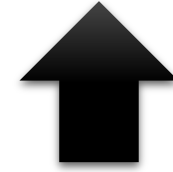
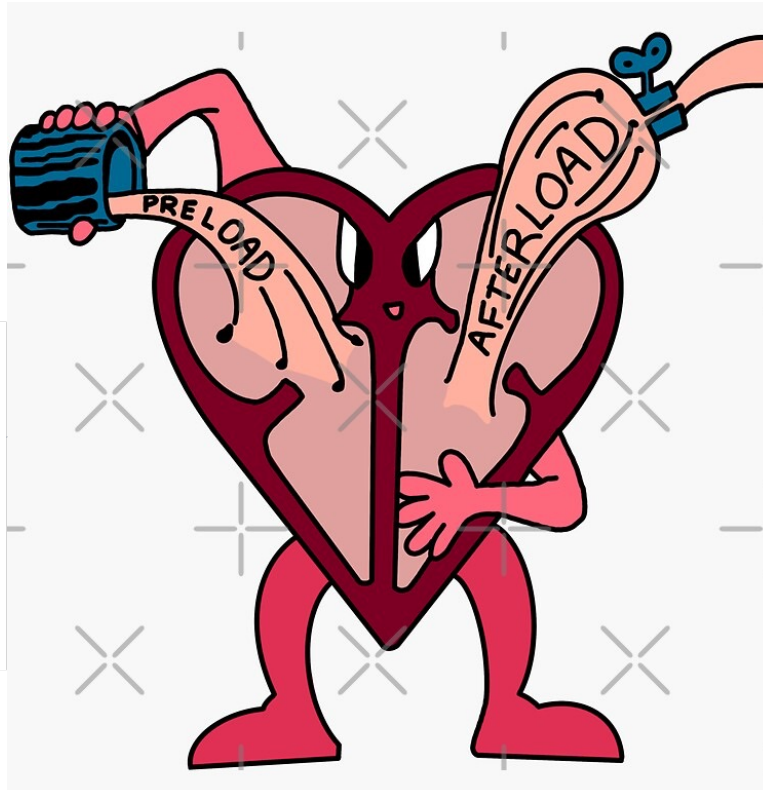
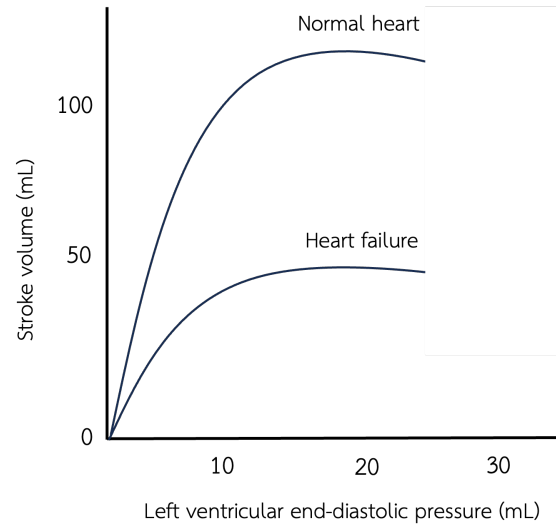
Pathophysiology



Preload

Blood volume filling in diastole phase

Frank-Starling law



Afterload

(Resistant)

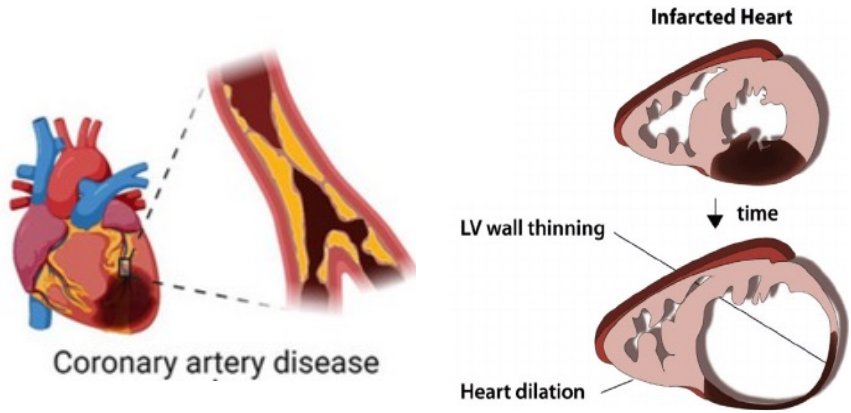
Left ventricular hypertrophy



Ventricular dysfunction

- Systolic dysfunction
- Diastolic dysfunction

Systolic dysfunction



Myocardial infarction

- LV dilatation
- Decrease cardiac pumping

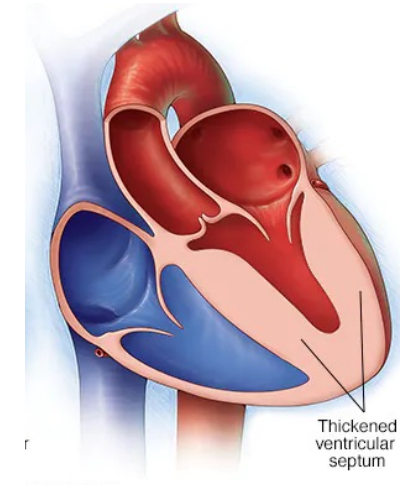
Ventricular dysfunction

Congestion

↓ Stroke volume

↓ Cardiac output

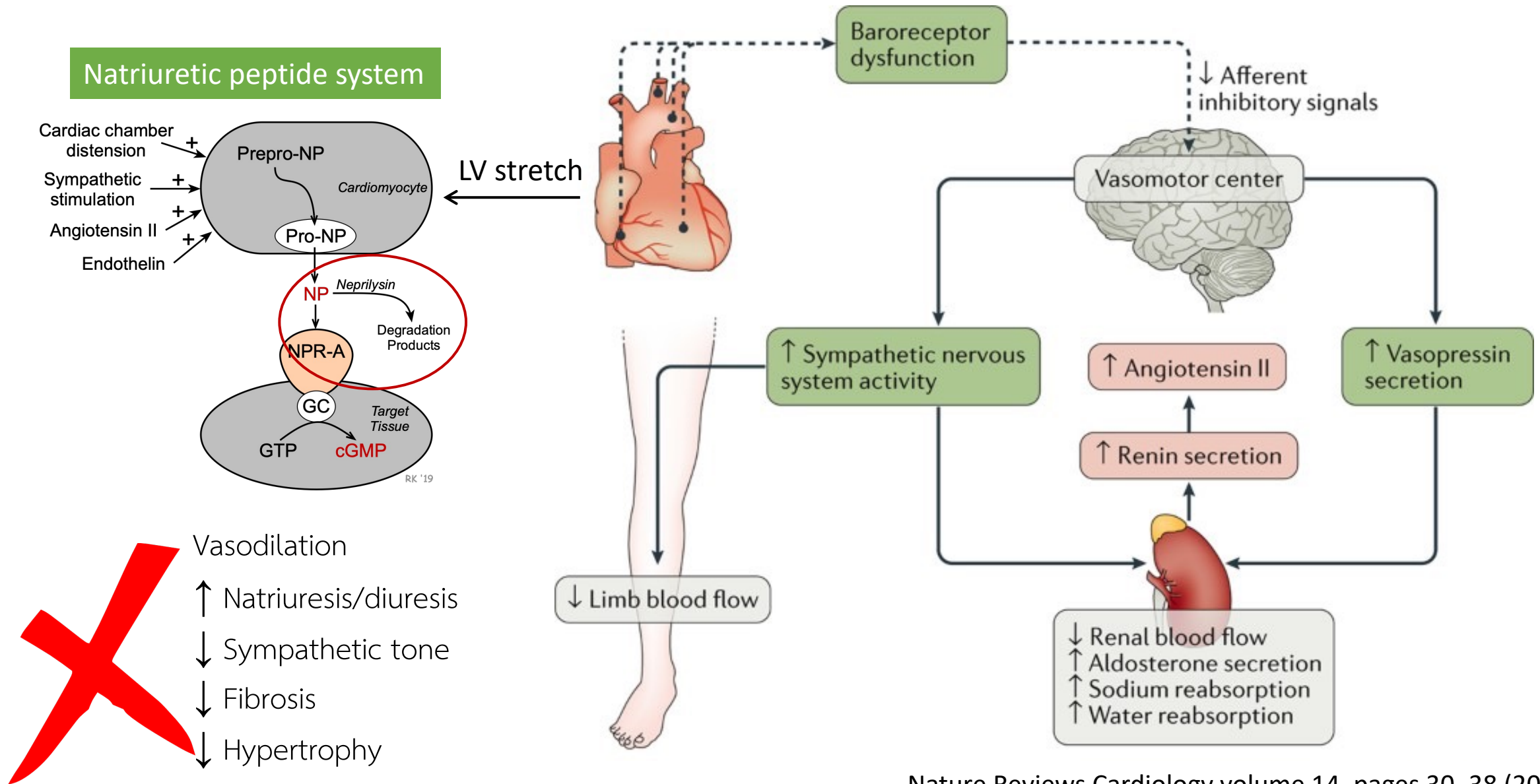
Diastolic dysfunction



Left ventricular hypertrophy

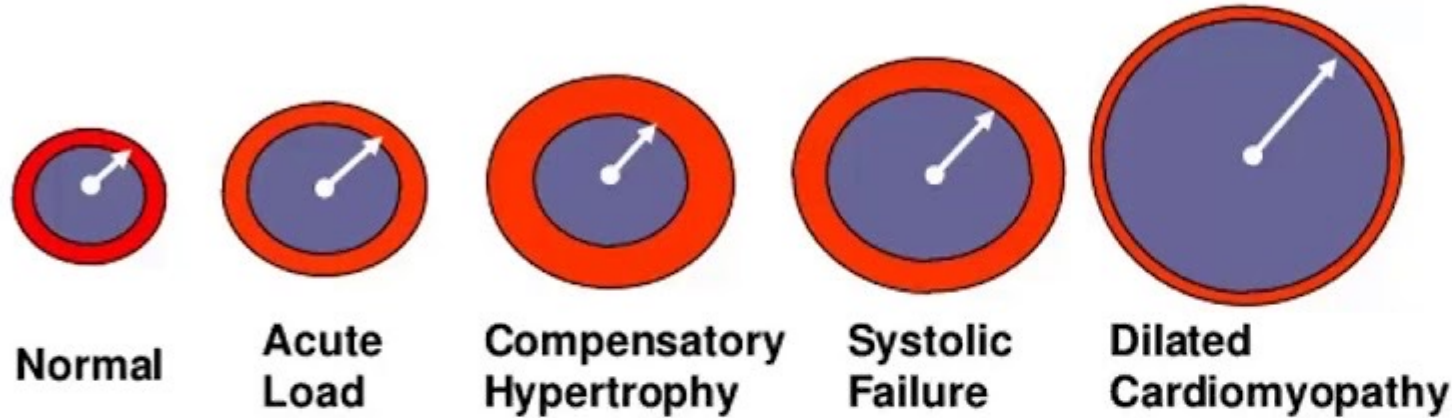
- Normal contraction
- Impaired ventricular compliance
- Impaired LV filling

Neurohormonal activation



La Place law of the heart

Pressure and Volume Govern Cardiac Function



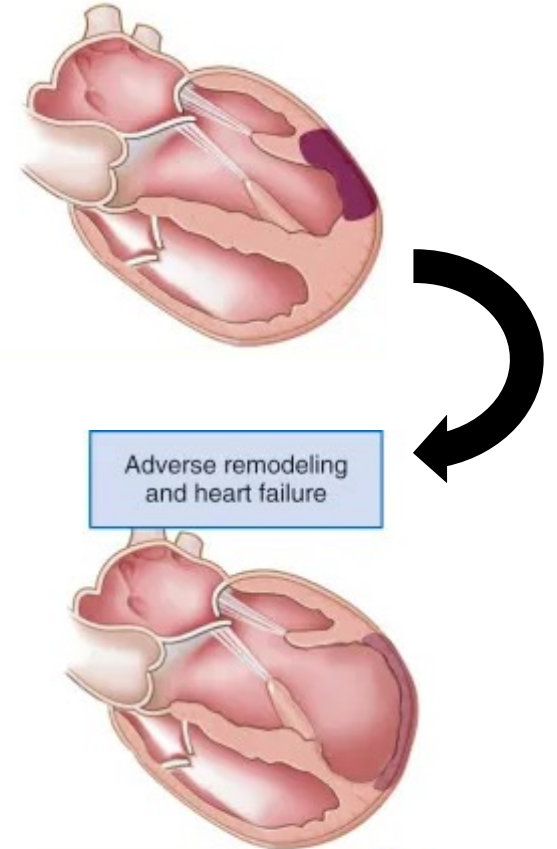
Primary Target of Heart Failure Therapy: Reduce LV Wall Stress

Laplace's Law: Wall stress = $\frac{\text{Pressure} \times \text{Radius}}{2 \times \text{Wall Thickness}} = \frac{\text{Pressure} \times \text{Volume}}{\text{LV Mass}}$

Wall Stress

La Place law for myocardial O₂ demand

$$VO_2 \propto \text{wall tension}$$



Type of HF

Rt. side Heart failure

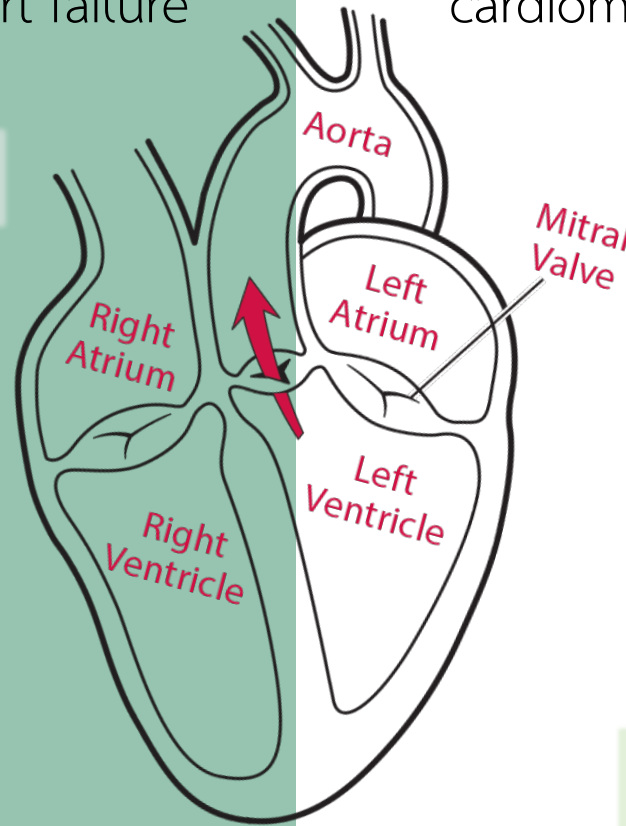
- **Etiology:** RV infarction, RV cardiomyopathy, pulmonary hypertension, Lt side heart failure

Congestion of peripheral tissues

- Jugular venous distention
- Peripheral edema

Liver congestion
Ascites

Gastrointestinal tract congestion
anorexia, weight loss



Valvular disease

Lt. side Heart failure

- **Etiology:** Ischemic cardiomyopathy, dilated-cardiomyopathy, valvular heart disease

Systolic dysfunction

Diastolic dysfunction

- Impaired contractility
- Increase afterload

Impaired
diastolic filling

Decrease cardiac output

Pulmonary
congestion

Dyspnea on exertion
Decrease tissue perfusion

Orthopnea
PND
Pink frothy sputum

ชายอายุ 70 ปี เหนื่อยมากขึ้นมา 3 วัน

- 1 สัปดาห์ก่อน เหนื่อยมากขึ้นตอนเดินในบ้าน เดินไปห้องน้ำแล้วเหนื่อย เดินขึ้นบันไดชั้น 2 ไม่ไหว ก้มผูกเชือกกรองเท้าแล้วเหนื่อย
- 3 วันก่อน นอนราบไม่ได้ ต้องหนุนหมอน 3 ใบ บางครั้งต้องลุกขึ้นมานั่ง อาการเหนื่อยจะดีขึ้น สังเกตว่าขา 2 ข้างบวมมากขึ้น

โรคประจำตัว โรคความดันโลหิตสูง 10 ปี
และโรคกล้ามเนื้อหัวใจขาดเลือด 1 เดือนก่อน

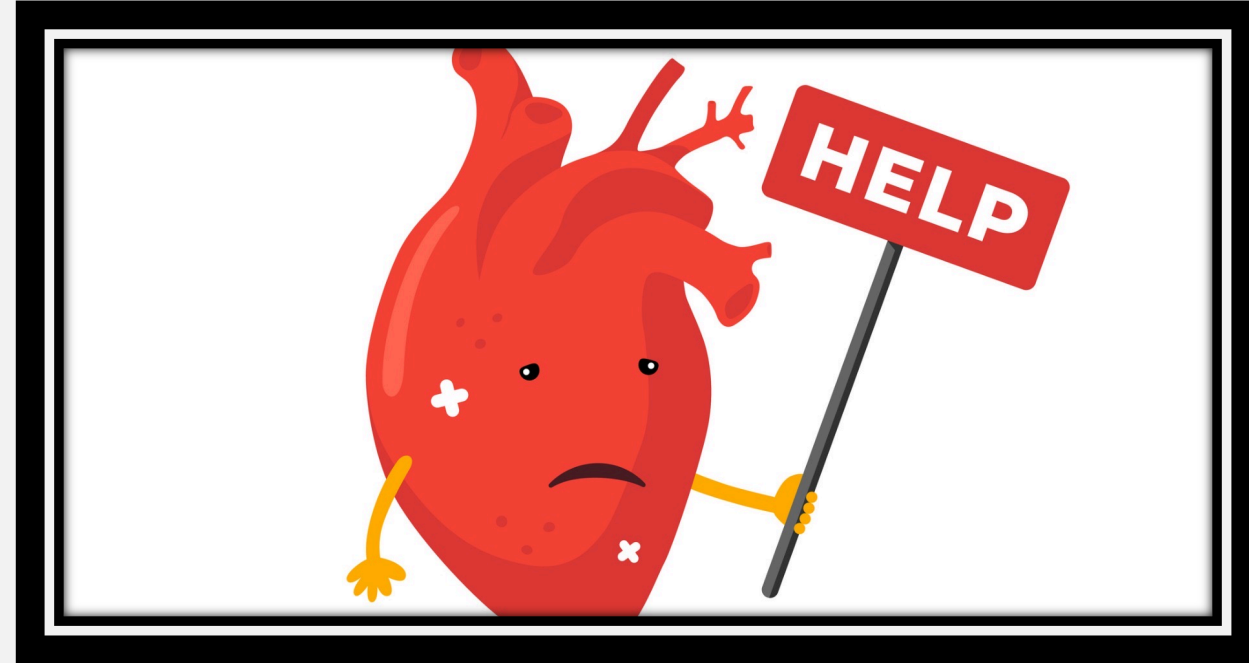


ตรวจร่างกายที่ห้องฉุกเฉิน

- General appearance: looked dyspnea
- V/S: BP 120/80 mmHg, PR 100/min, regular pulse, RR 30/min
- JVP 5 cmH₂O (upright 45°)
- Heart: PMI at 6th ICS, mid-clavular line, no murmur, S3 gallop
- Lungs: fine crepitation and wheezing both lungs
- Extremities: pitting edema 2+



Clinical presentation



Typical symptoms and signs

Symptoms: Dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea, reduced exercise tolerance, fatigue, tiredness, increased time to recover after exercise, ankle swelling

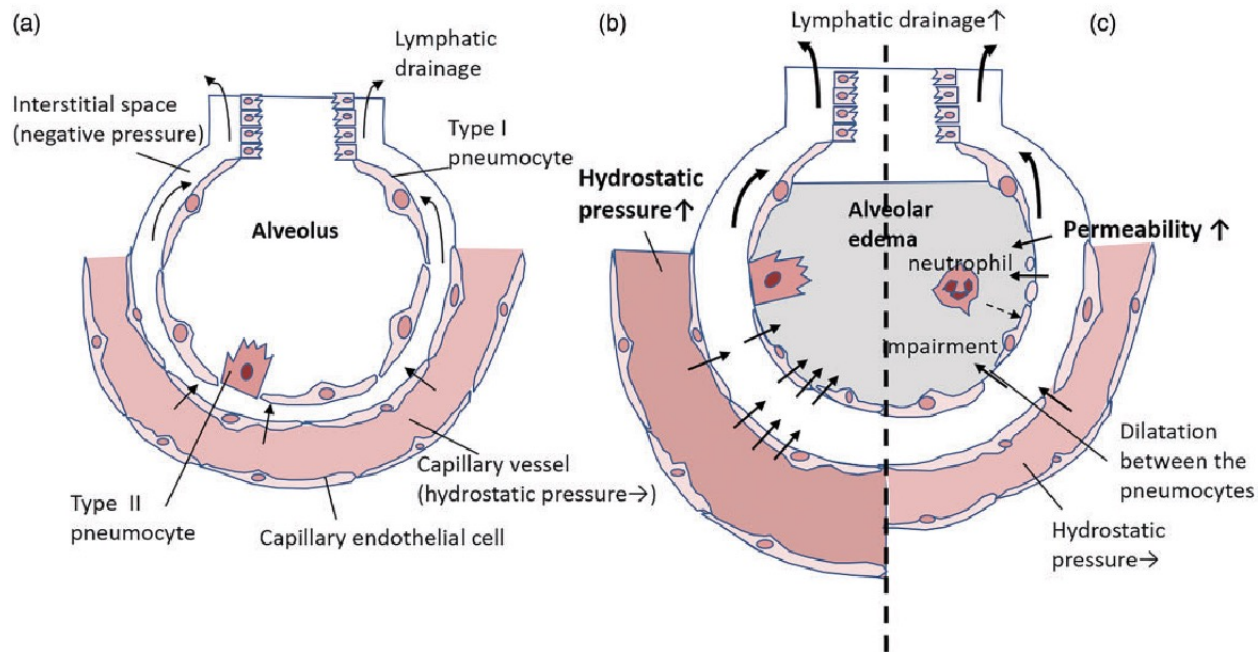
Signs: Elevated jugular venous pressure, hepatojugular reflux, third heart sound (gallop rhythm), laterally displaced apical impulse

Less typical and less specific

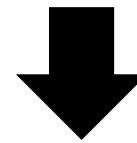
Symptoms: Nocturnal cough, wheezing, bloated feeling, loss of appetite, confusion (especially in the elderly), depression, palpitation, dizziness, syncope, bendopnea, Peripheral oedema (ankle, sacral, scrotal), pulmonary crepitations, pleural effusion

Orthopnea is shortness of breath (dyspnea) that happens when you're lying on your back (in a supine position). Sitting or standing up relieves this symptom.

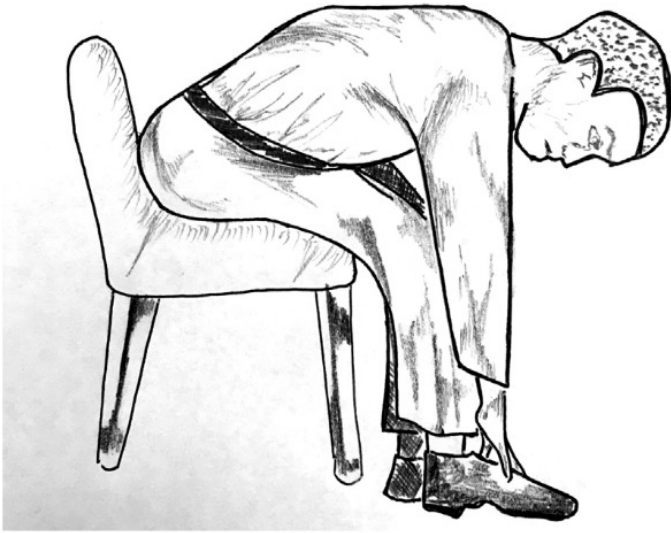
Paroxysmal nocturnal dyspnea is a sensation of shortness of breath that awakens the patient, often after 1 or 2 hours of sleep, and is usually relieved in the upright position.



Horizontal position/Lying down



Redistribution of blood volume
from the lower extremities and
splanchnic beds to the lungs

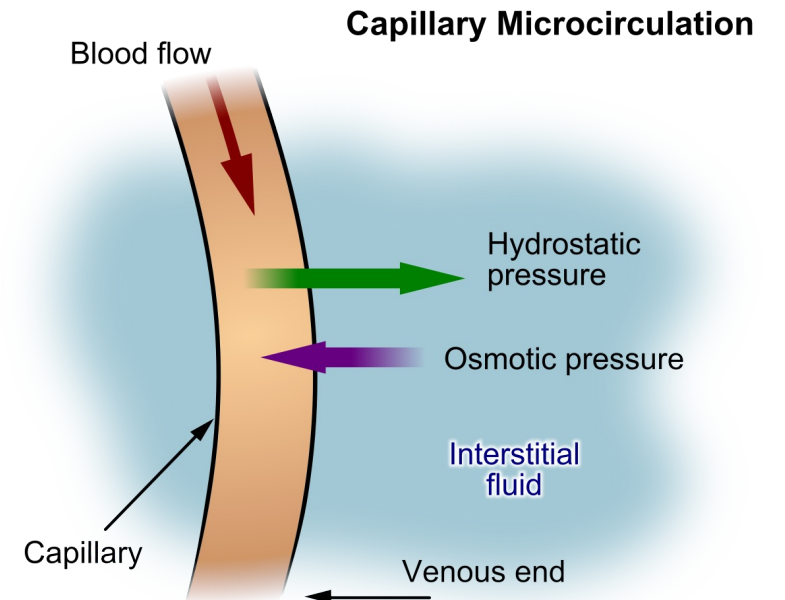


Bendopnea

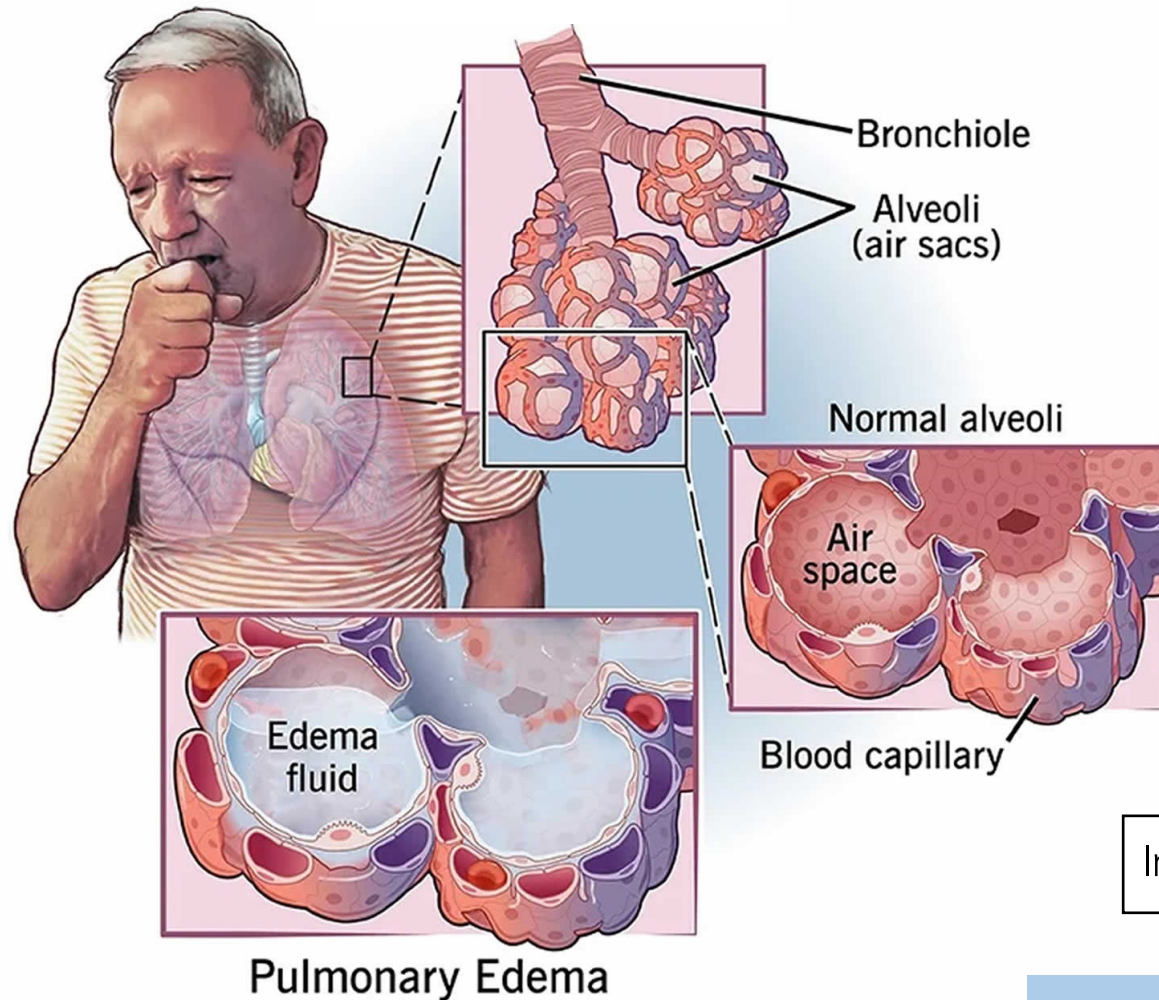
A patient sits in a chair, bends at the waist, and touches his feet. If present of dyspnea within 30s of bending.



Pitting edema is a type of swelling, often in the lower extremities, that causes enough fluid retention to leave a pit (or indentation).



Pink Frothy sputum



Acute pulmonary edema

Increase LV pressure



Increase pulmonary venous pressure

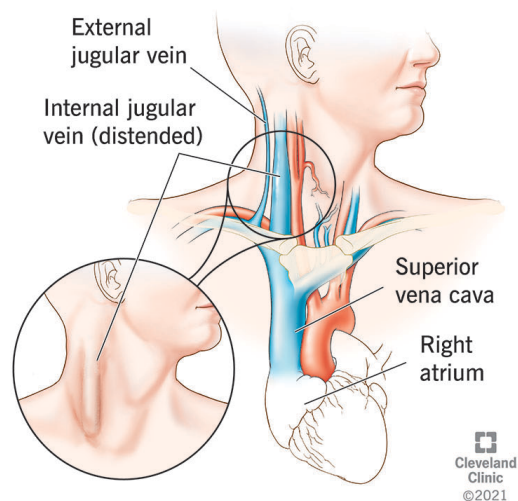
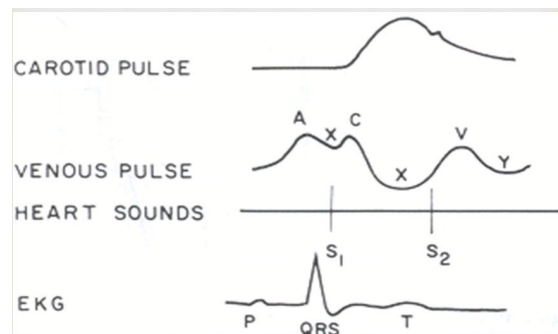


Increase lung capillary pressure and hydrostatic pressure



Blood and fluids leak in the tissues and alveolar sacs

Jugular venous pressure



Reflects right atrial pressure (central venous pressure = JVP + 5 cm), is estimated by examining the internal jugular veins.

- Do not recommend using the external jugular vein pulsations to estimate central venous pressure, because valves in these veins may lead to inaccurate readings.

Hepatojugular reflux



<https://geekymedics.com/jugular-venous-pressure-jvp/>

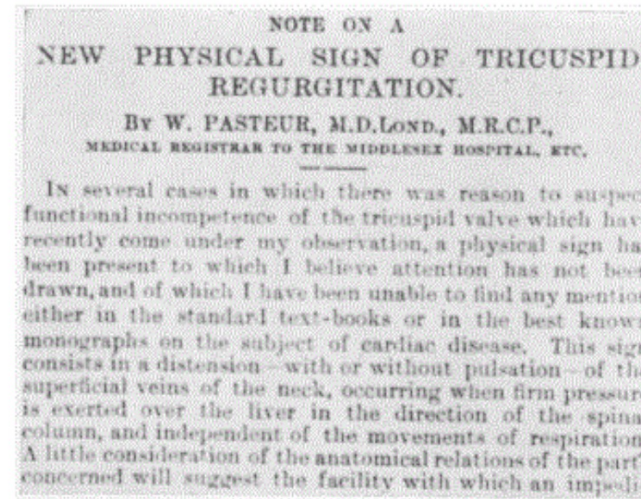


Fig. 1.—The hepatojugular reflux was originally reported as a manifestation of tricuspid regurgitation by William Pasteur.

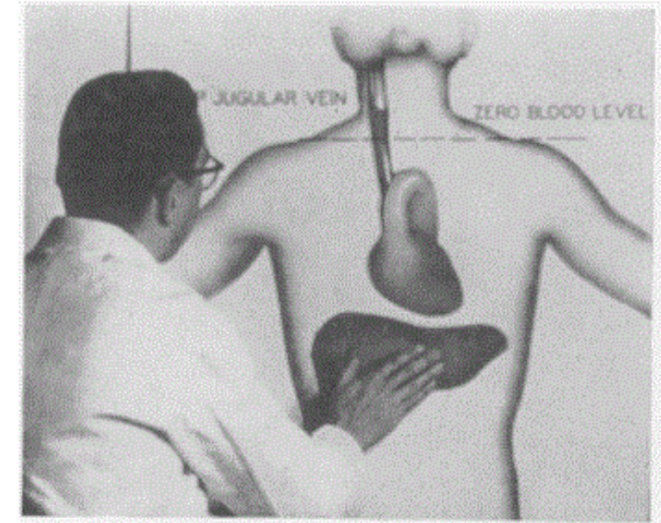
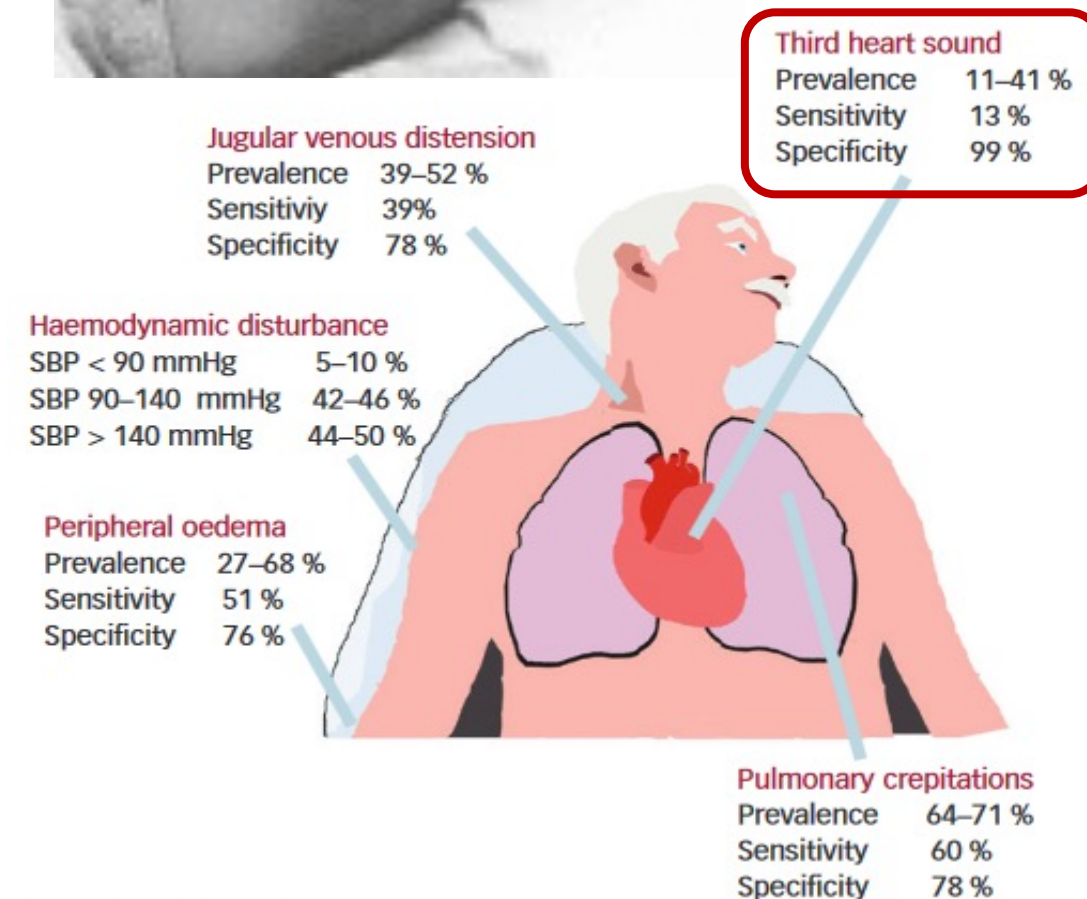
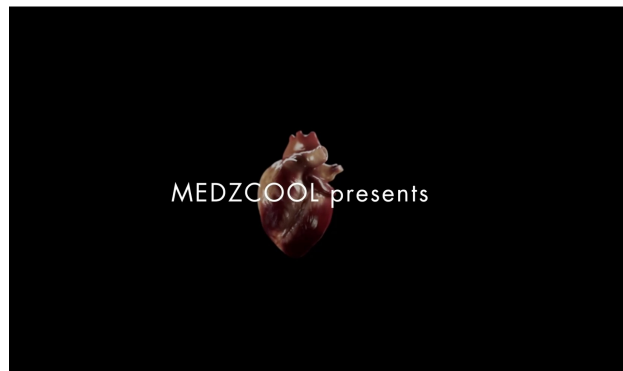


Fig. 2.—This sign consists of an elevation of the pulsating level in the deep neck veins or distention of the superficial neck veins when pressure is exerted over the liver.

The hepatojugular reflux, as presently defined, consists of a distention of the neck veins when pressure is applied over the liver. With a competent heart, pressure on the liver does not elevate the venous blood level in the neck veins when the subjects are in the semirecumbent position.

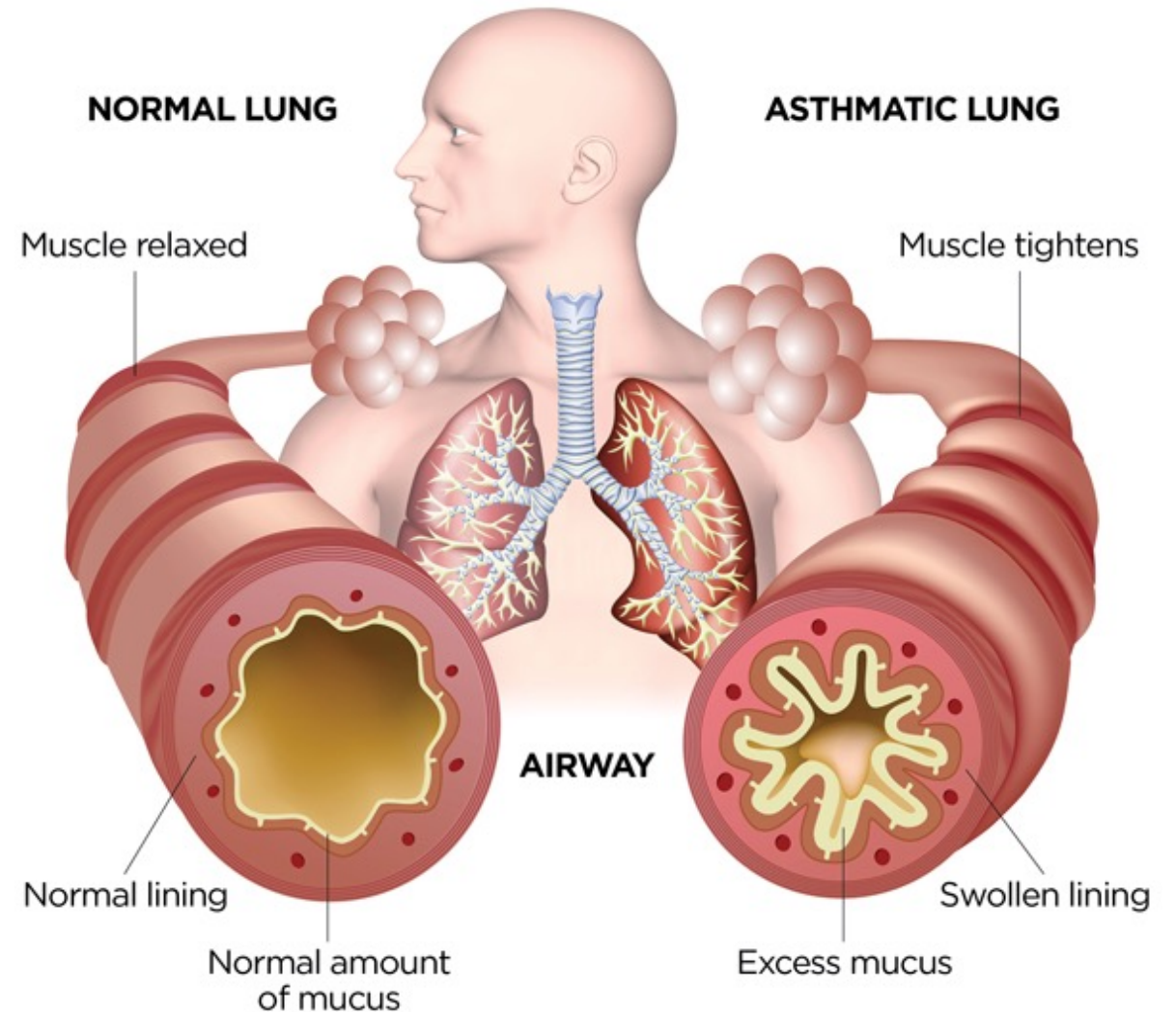
S3 gallop

- The patient is examined supine and then turned to a 30° left lateral position with the left arm extended upward away from the chest.
- The left lateral position is of critical importance because the ventricular gallop is often heard only with the patient turned to the side.
- After the apical impulse is located by careful palpation, the bell of the stethoscope is placed lightly over the apex.



Cardiac wheezing/cardiac asthma

- Difficult breathing due to cardiogenic pulmonary edema.
- The pathogenesis is not asthmatic bronchoconstriction but rather a reflex bronchoconstriction as a manifestation of pulmonary congestion due to pulmonary venous hypertension.



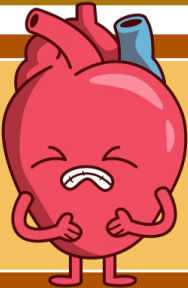
New York Heart Association (NYHA) Functional Class



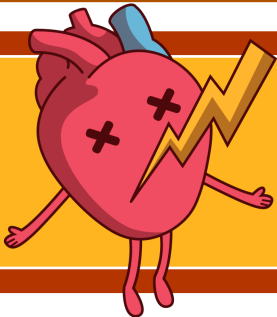
NYHA functional class I: ผู้ป่วยไม่มีอาการใด ๆ สามารถทำกิจกรรมปกติ (ordinary physical activity) ได้โดยไม่มีอาการหายใจลำบาก หอบเหนื่อย



NYHA functional class II: ผู้ป่วยมีข้อจำกัดบ้างเพียงเล็กน้อย (slight limitation) ในการทำกิจกรรมปกติโดยผู้ป่วยมักมีอาการเหนื่อยเมื่อทำกิจกรรมที่ต้องออกแรงมาก ๆ

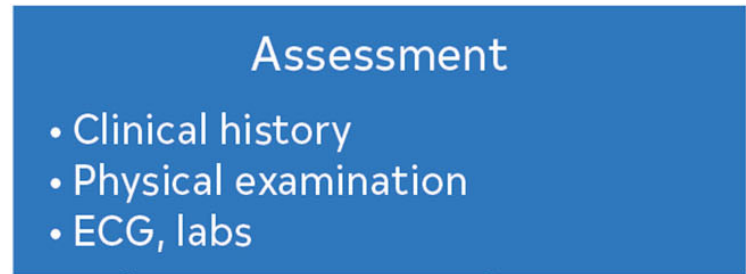


NYHA functional class III: ผู้ป่วยมีข้อจำกัดมากพอสมควรในการทำกิจกรรมปกติ โดยมีอาการหายใจลำบาก หรือ หอบเหนื่อยอย่างรวดเร็วเมื่อทำกิจวัตรประจำวัน เช่น อาบน้ำแต่งตัว แต่จะไม่มีอาการขณะพัก (no symptoms at rest)

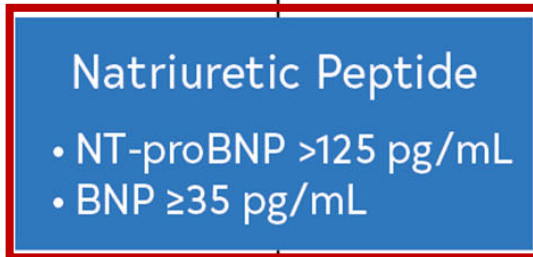
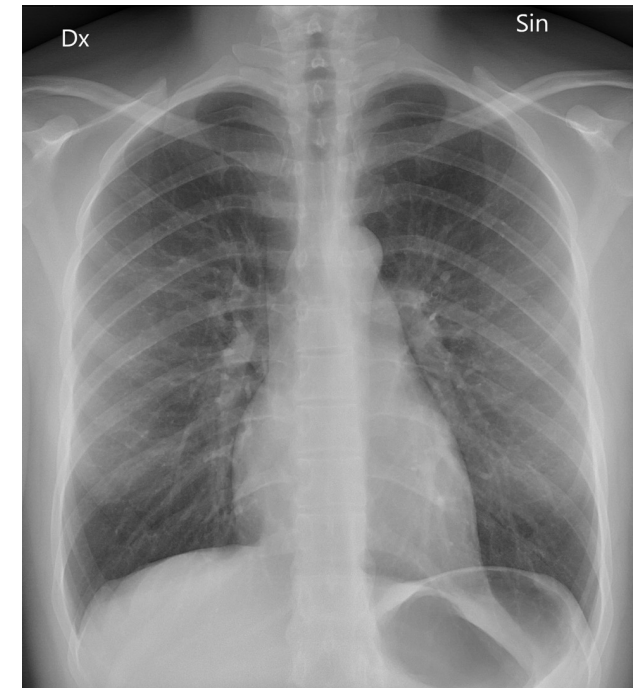


NYHA functional class IV: ผู้ป่วยมีข้อจำกัดอย่างมากในการทำกิจกรรมปกติ มีอาการเหนื่อย หอบขณะพัก (symptoms at rest)

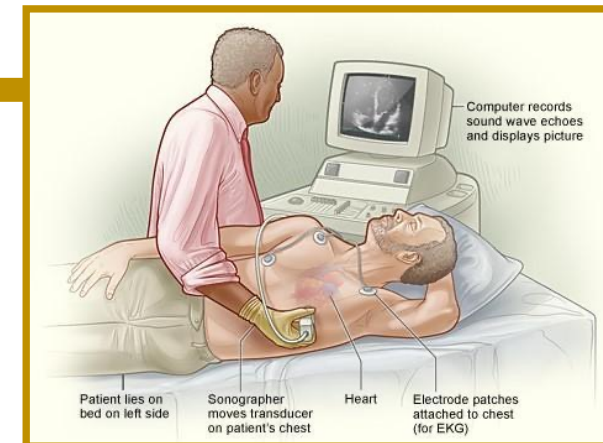
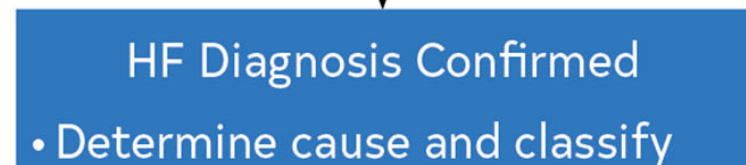
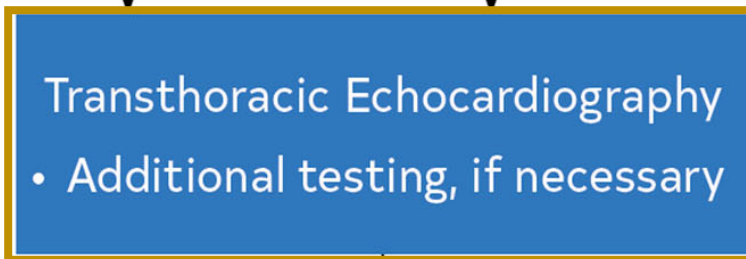
Diagnostic Algorithm for Patients With Suspected HF



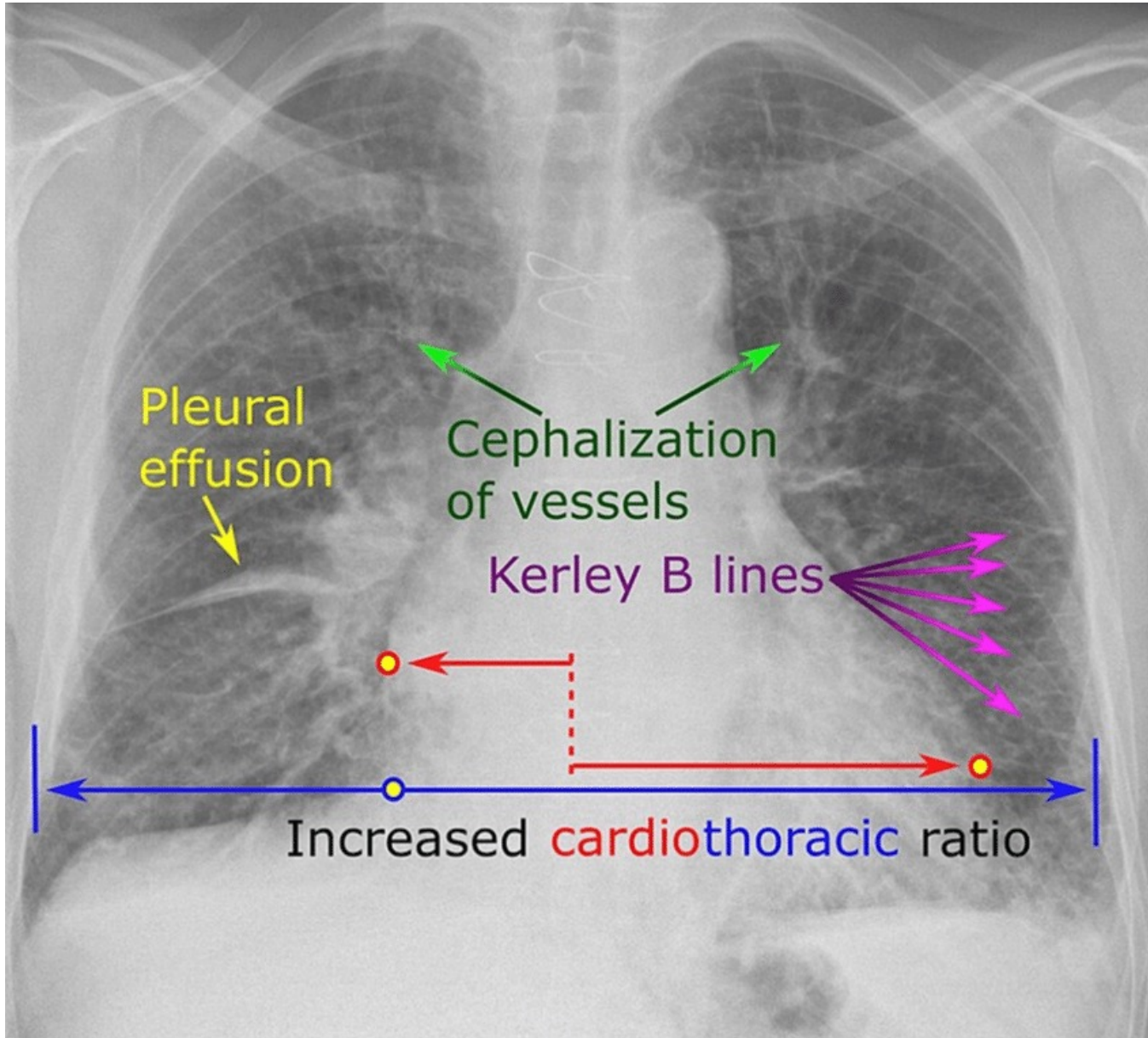
+ CXR



Cardiac biomarker



Chest X-ray in patient with heart failure



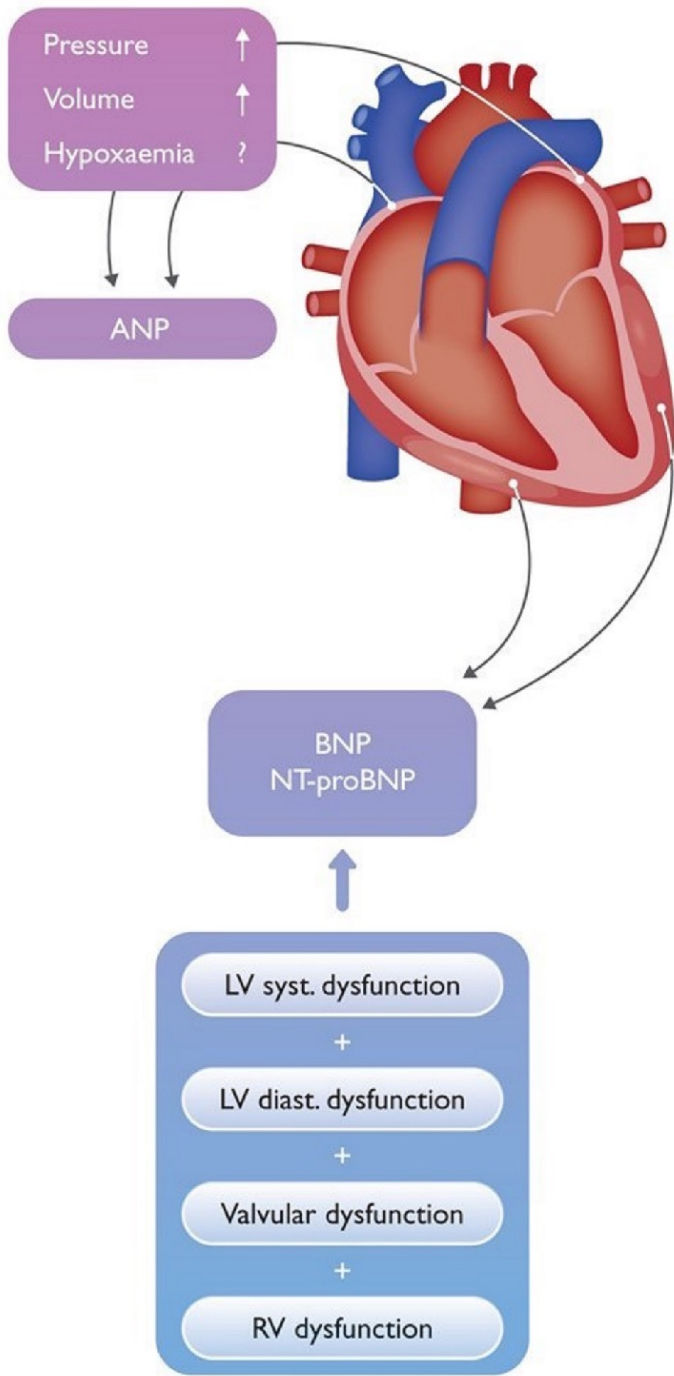
A: alveolar opacification

B: bat winging

C: cardiomegaly

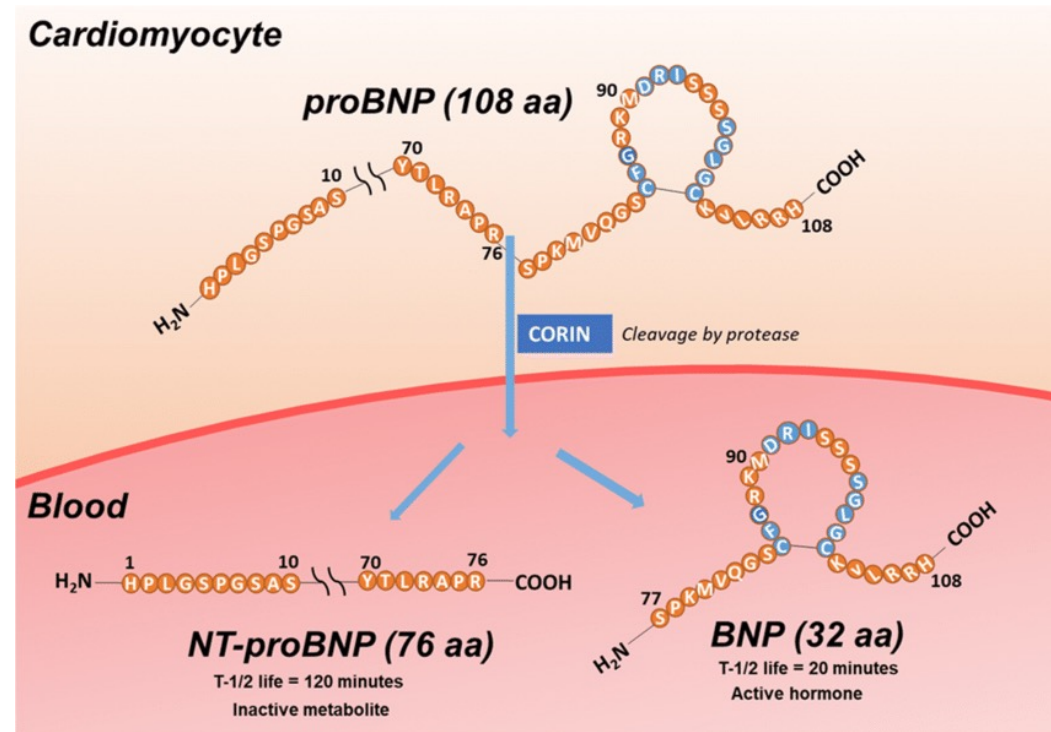
D: diffuse interstitial thickening
(septal lines) and diversion (vascular
upper zone diversion, cephalization)

E: effusions (pleural)



NT-proBNP (N-terminal pro-B-type natriuretic peptide)

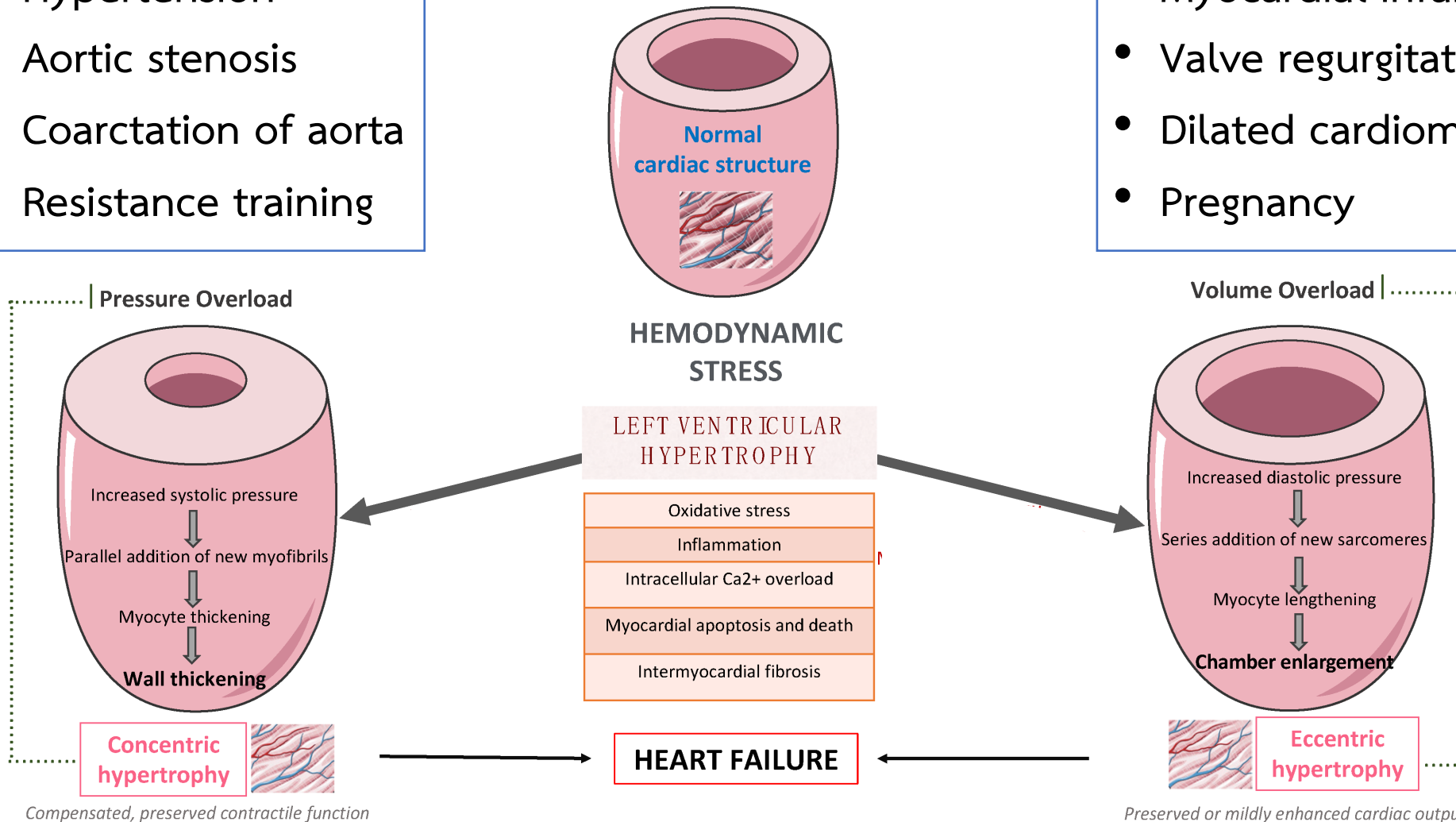
is the N-terminal fragment of the B-type natriuretic peptide, secreted by myocytes as a reaction to several stimuli including wall stretch.



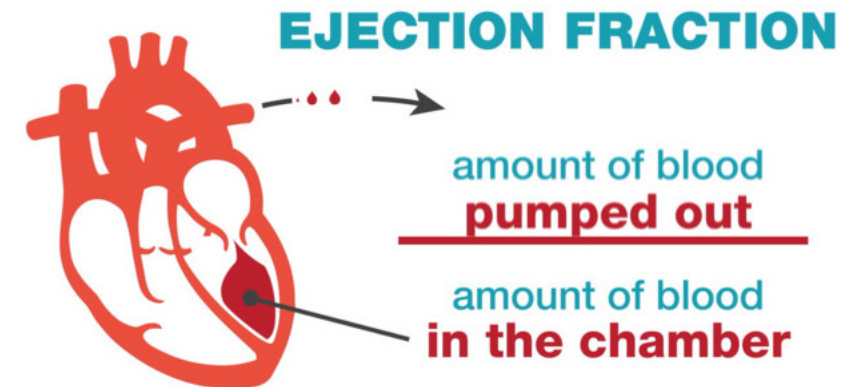
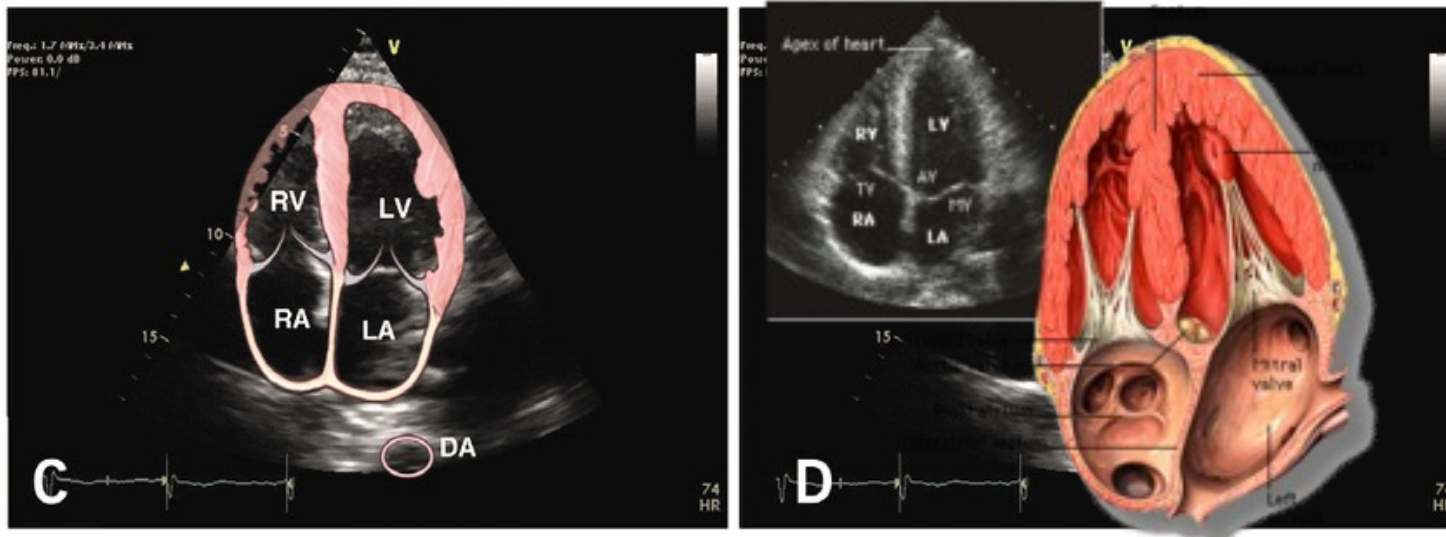
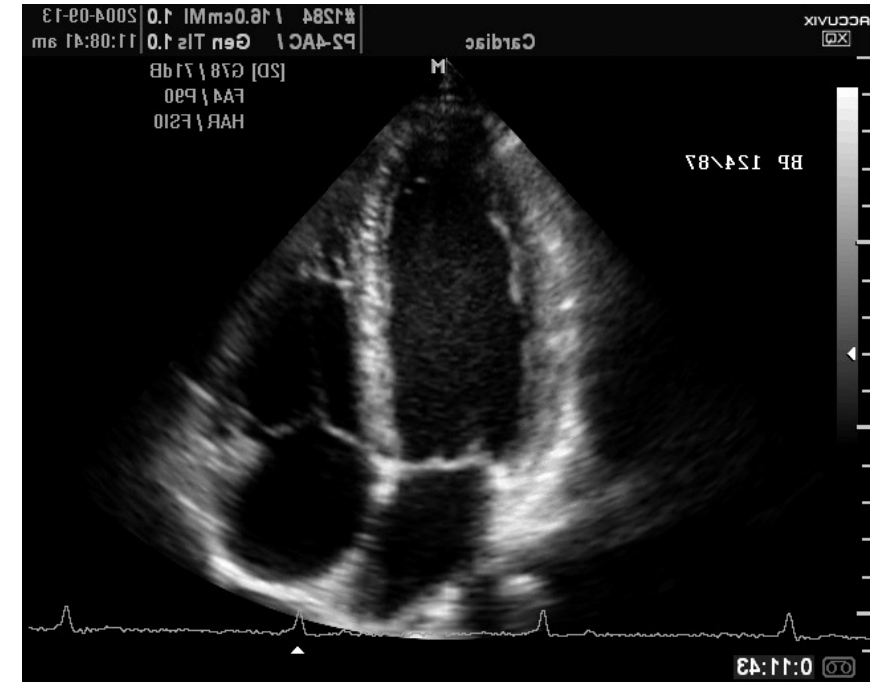
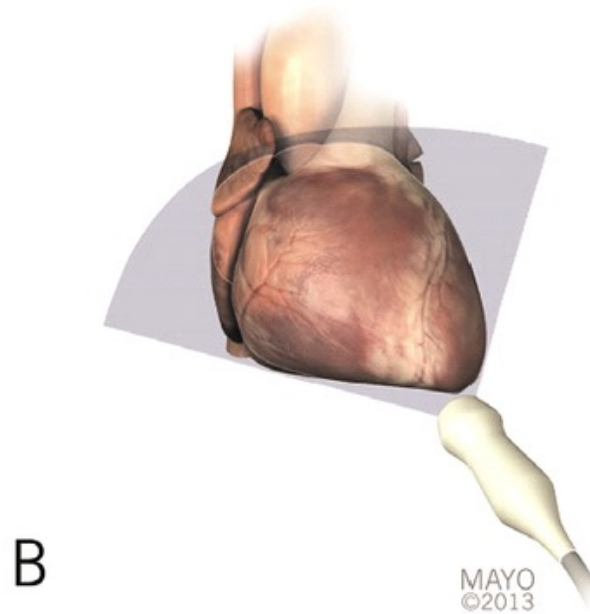
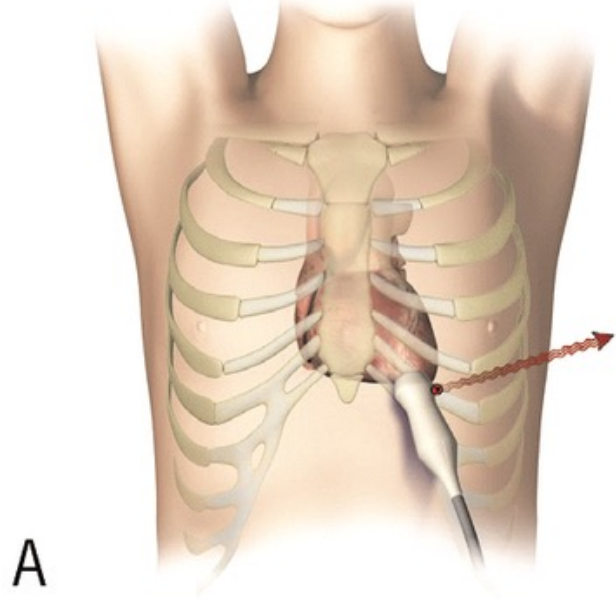
Etiology

- Hypertension
- Aortic stenosis
- Coarctation of aorta
- Resistance training

- Myocardial infarction
- Valve regurgitation
- Dilated cardiomyopathy
- Pregnancy



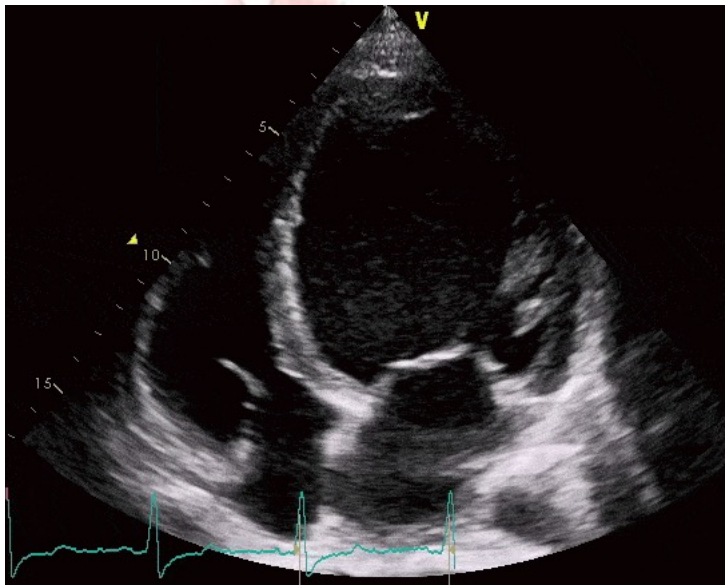
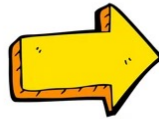
Transthoracic echocardiogram



Classification of HF by LVEF



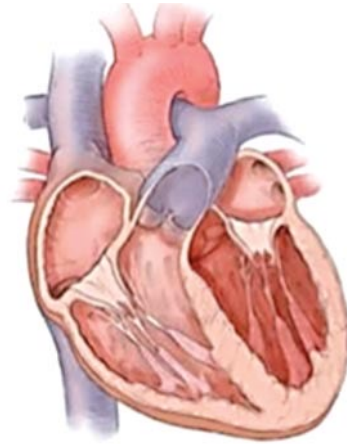
Heart failure reduced EF (HFrEF)



- Enlarged left ventricle
- Thinned walls
- Reduced pumping ability
- Weakened muscle

LVEF $\leq 40\%$

Heart failure improved EF (HFimpEF):
Previous LVEF $\leq 40\%$ and a follow-up measurement of LVEF $> 40\%$



Heart failure mildly reduced EF (HFmpEF):
LVEF 41%–49% with evidence of spontaneous or provokable increased LV filling pressures

Heart failure preserved EF (HFpEF)



LVEF $\geq 50\%$ with evidence of spontaneous or provokable increased LV filling pressures

ชายอายุ 70 ปี
เหนื่อยมากขึ้นมา 3 วัน

- 1 สัปดาห์ก่อน เหนื่อยมากขึ้นตอนเดินในบ้าน เดินไปห้องน้ำแล้วเหนื่อย เดินขึ้นบันไดชั้น 2 ไม่ไหว ก้มผูกเชือกกรองเท้าแล้วเหนื่อย
- 3 วันก่อน นอนราบไม่ได้ ต้องหนุนหมอน 3 ใบ บางครั้งต้องลุกขึ้นมานั่ง อาการเหนื่อยจะดีขึ้น สังเกตว่าขา 2 ข้างบวมมากขึ้น

โรคประจำตัว โรคความดันโลหิตสูงและ
โรคกล้ามเนื้อหัวใจขาดเลือด





Treatment in the patients

Acute treatment for acute HF

1

Treat congestion+/-
hypoperfusion

Congestion

Perfusion

Warm and Dry	Warm and Wet
Cold and Dry	Cold and Wet

Congestion(Wet)

- Orthopnea/PND
- Jugular venous distention
- Lower extremities edema
- Hepatojugular reflux
- Ascites

Hypoperfusion(Cold)

- Hypotension, narrow pulse pressure
- Cold extremities
- Altered mental status
- Oliguria

2

Identify precipitating factors
and comorbidities

- C** acute Coronary syndrome
- H** Hypertension emergency
- A** Arrhythmia
- M** Mechanical cause
- P** Pulmonary embolism
- I** Infection
- T** Tamponade

Acute treatment for acute HF

- Up titration of guideline-directed medical therapy (GDMT)
- Treat comorbidity

Warm and Dry

Warm and Wet

- Diuretics
- Vasodilators
- Consider renal replacement therapy

Cold and Dry

Cold and Wet

- Fluid challenge test
- Inotropic agent

- Inotropic agent
- Diuretics
- Consider Vasopressor if refractory hypotension
- Mechanical support

Case scenario

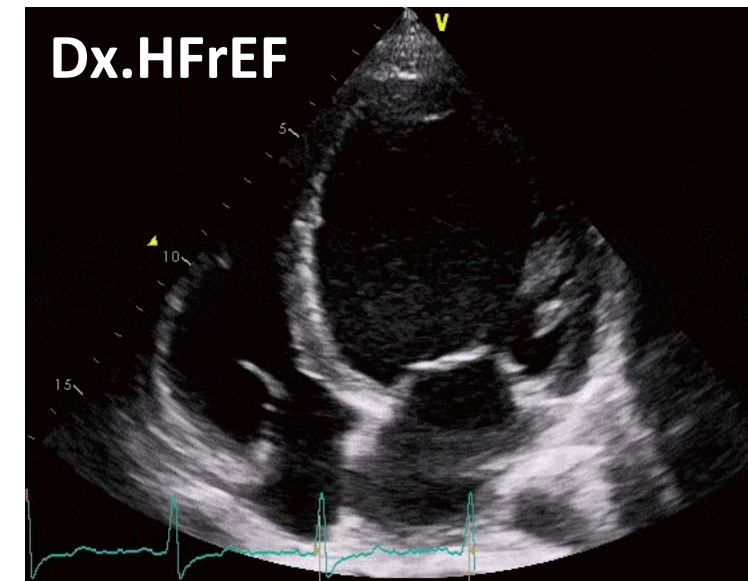
Physical examination

- V/S: BP 120/80 mmHg, PR 100/min, regular pulse, RR 30/min
- JVP 5 cmH₂O (upright 45°)
- Heart: PMI at 6th ICS, mid-clavular line, no murmur, S3 gallop
- Lungs: fine crepitation both lungs
- Extremities: pitting edema 2+

Warm and Dry	Warm and Wet
Cold and Dry	Cold and Wet



Management : Diuretics



LVEF 20%

Guideline-directed medical therapy according to LVEF

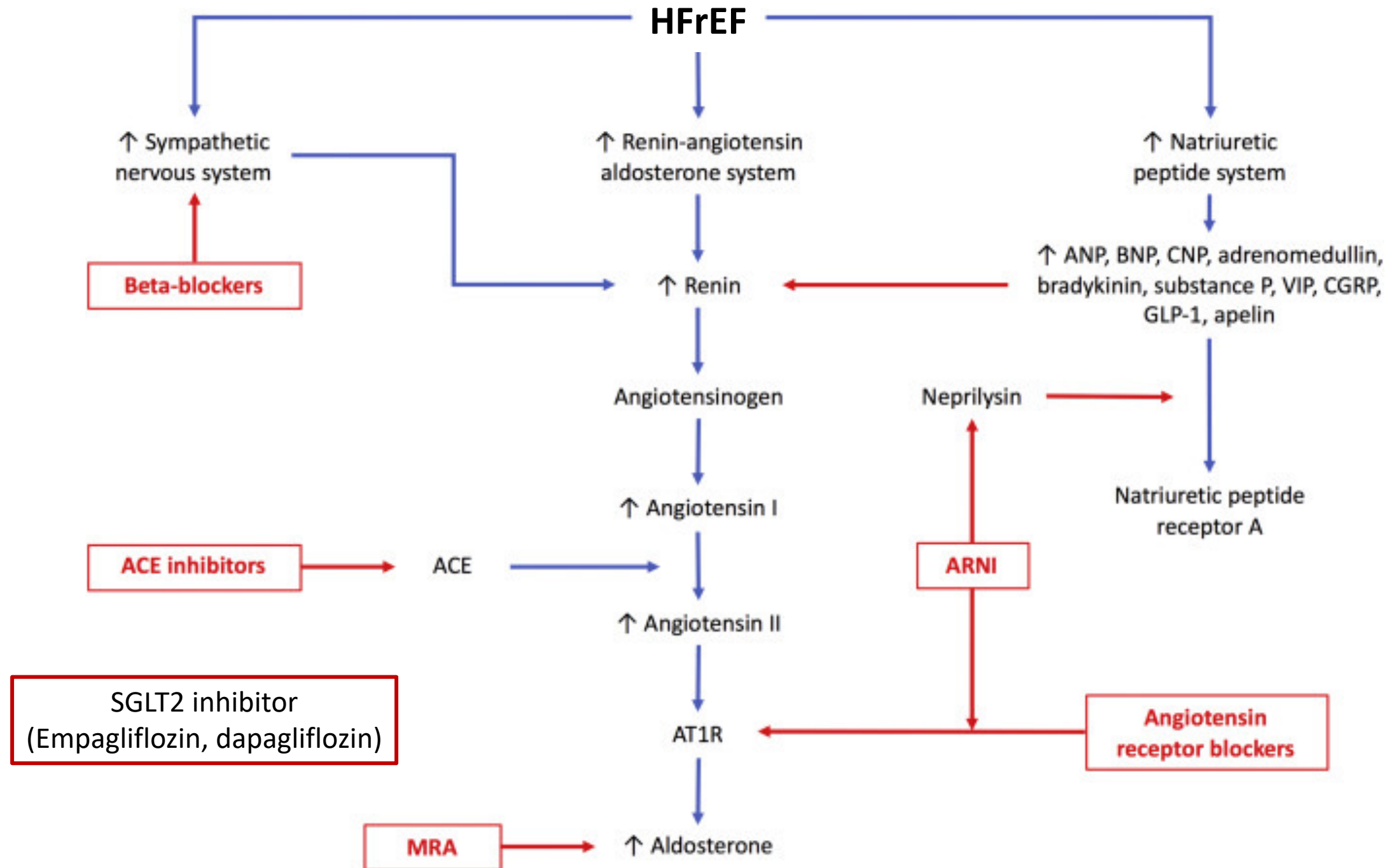
HFrEF

- ACEI/ARNI
- Beta-blocker
- MRA
- SGLT2i (Dapagliflozin/Empagliflozin)
- Loop diuretic for fluid retention

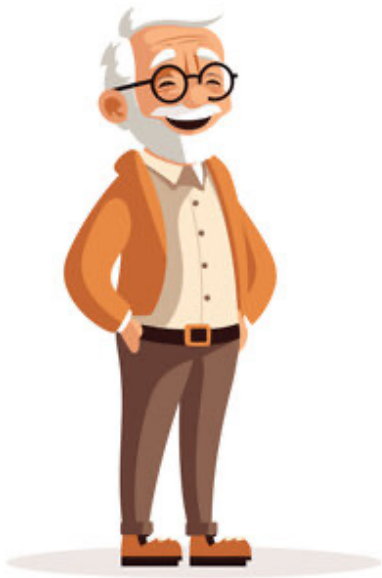
HFimpEF

HFpEF

- SGLT2i (Dapagliflozin/Empagliflozin)
- Loop diuretic for fluid retention



Summary



Dx. Acute heart failure – HFrEF

Cause: CHAMP-IT

Management: diuretic

neurohormonal blocking agent

- ACEIs/ARBs/ARNI
- Beta-blocker
- MRA
- SGLT2i (Dapagliflozin/Empagliflozin)
- Loop diuretic for fluid retention

Q & A



แบบฟอร์มประเมินอาจารย์



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Audience Q&A

① Start presenting to display the audience questions on this slide.